

STUDYHUB
CLASS 12 — MP BOARD
MATHEMATICS
ORIGINAL SAMPLE QUESTION PAPER — SET 1

Time: 3 Hours	Maximum Marks: 80
Medium: English	Set: 1

Note: This is an original StudyHub practice paper based on the MPBSE syllabus and examination pattern. It is not an official MPBSE paper.

GENERAL INSTRUCTIONS

1. All questions are compulsory.
2. Questions 1 to 5 contain objective-type sub-questions.
3. Questions 6 to 15 carry 2 marks each.
4. Questions 16 to 19 carry 3 marks each.
5. Questions 20 to 23 carry 4 marks each.
6. Internal choices are given where indicated.
7. Use of calculator is not permitted.

SECTION - A : OBJECTIVE TYPE QUESTIONS

Q.1 Choose and write the correct option. ($1 \times 6 = 6$)

(i) If $A = \{1, 2, 3\}$, the number of reflexive relations on A is:

- A. 3
- B. 6
- C. 8
- D. 9

(ii) The principal value of $\cos^{-1}(-1)$ is:

- A. 0
- B. $\pi/2$
- C. π
- D. 2π

(iii) If A is a square matrix of order 2 and $|A| = 5$, then $|3A|$ is:

- A. 15
- B. 30
- C. 45
- D. 90

(iv) If $f(x) = x^3 - 3x$, then $f'(1)$ is:

- A. -3
- B. 0
- C. 3
- D. 6

(v) The magnitude of $3i + 4j$ is:

- A. 3
- B. 4
- C. 5
- D. 7

(vi) If A and B are independent events, then:

- A. $P(A \cap B) = P(A) + P(B)$
- B. $P(A \cap B) = P(A)P(B)$
- C. $P(A \cap B) = P(A) - P(B)$
- D. $P(A \cap B) = 0$

Q.2 Fill in the blanks. ($1 \times 6 = 6$)

(i) A relation which is both symmetric and transitive but not necessarily reflexive is called a _____ relation.

(ii) $d/dx(\sin x) =$ _____.

(iii) If A is a singular matrix, then $|A| =$ _____.

(iv) The integrating factor of $dy/dx + P(x)y = Q(x)$ is _____.

(v) The dot product of two perpendicular vectors is _____.

(vi) The probability of an impossible event is _____.

Q.3 Write True or False. ($1 \times 6 = 6$)

(i) Every one-one function is necessarily onto.

(ii) $f(x) = |x|$ is continuous at $x = 0$.

(iii) If A is an identity matrix, then $A^{-1} = A$.

- (iv) The derivative of x^2 is $2x$.
 (v) Two parallel non-zero vectors have zero cross product.
 (vi) $P(A') = 1 - P(A)$.

Q.4 Match Column A with Column B. ($1 \times 7 = 7$)

Column A	Column B
(i) $\int (1/x) dx$	(a) $\tan^{-1}x + C$
(ii) $\int 1/(1+x^2) dx$	(b) $\log x + C$
(iii) $d/dx(e^x)$	(c) e^x
(iv) $d/dx(\log x)$	(d) $1/x$
(v) $\vec{a} \cdot \vec{a}$	(e) $ \vec{a} ^2$
(vi) $P(A \cup B)$	(f) $P(A) + P(B) - P(A \cap B)$
(vii) $\int \cos x \, dx$	(g) $\sin x + C$

Q.5 Answer in one word/one sentence. ($1 \times 7 = 7$)

- (i) Write the order of $d^2y/dx^2 + 3dy/dx + y = 0$.
 (ii) Find the minimum value of $x^2 + 4$.
 (iii) Find the modulus of $6i - 8j$.
 (iv) What is the probability of a sure event?
 (v) Find $\sin^{-1}(0)$.
 (vi) If $\vec{a} \cdot \vec{b} = 0$, what is the angle between two non-zero vectors?
 (vii) Write the derivative of $\tan^{-1}x$.

SECTION - B : VERY SHORT ANSWER QUESTIONS

Q.6 (2 Marks)

Prove that $R = \{(1,1), (2,2), (3,3), (1,2), (2,1)\}$ on $A = \{1,2,3\}$ is reflexive and symmetric.

OR

Show that $f: R \rightarrow R$, $f(x) = 2x + 3$ is one-one.

Q.7 (2 Marks)

Simplify $\tan^{-1}((1 - \cos x)/\sin x)$, $0 < x < \pi$.

OR

Find $\sin^{-1}(3/5) + \cos^{-1}(3/5)$.

Q.8 (2 Marks)

If $A = \begin{bmatrix} 2 & 1 \\ 3 & 4 \end{bmatrix}$, find A^{-1} .

OR

Find the determinant $|\begin{bmatrix} 3 & 2 \\ 5 & 4 \end{bmatrix}|$.

Q.9 (2 Marks)

Find the first and second derivatives of $y = x^3 - 5x^2 + 4x$.

OR

Find the equation of tangent to $y = x^2$ at $x = 2$.

Q.10 (2 Marks)

Find the intervals in which $f(x) = x^2 - 4x + 3$ is increasing and decreasing.

OR

Find the maximum value of $f(x) = 6x - x^2$.

Q.11 (2 Marks)

Evaluate $\int (3x^2 + 4x - 5) dx$.

OR

Evaluate $\int 2x/(x^2 + 1) dx$.

Q.12 (2 Marks)

Evaluate $\int_0^1 (2x + 1) dx$.

OR

Find the area bounded by $y = x$, the x-axis and $x = 2$.

Q.13 (2 Marks)

Solve $dy/dx = 3x^2$.

OR

Find the integrating factor of $dy/dx + 2y = e^x$.

Q.14 (2 Marks)

Find the angle between $\vec{a} = 2\mathbf{i} + \mathbf{j}$ and $\vec{b} = \mathbf{i} - 2\mathbf{j}$.

OR

Find a unit vector in the direction of $3\mathbf{i} + 4\mathbf{j}$.

Q.15 (2 Marks)

A die is thrown once. Find the probability of getting a prime number.

OR

Two coins are tossed simultaneously. Find the probability of getting exactly one head.

SECTION - C : SHORT ANSWER QUESTIONS

Q.16 (3 Marks)

Using matrix method, solve $2x+y=7$, $x-y=1$.

OR

Find the inverse of $A = \begin{bmatrix} 1 & 2 \\ 2 & 5 \end{bmatrix}$ using elementary operations.

Q.17 (3 Marks)

Find the local maximum and local minimum values of $f(x) = x^3 - 6x^2 + 9x + 2$.

OR

Find the points on $y = x^3 - 3x^2 + 2$ where the tangent is parallel to the x-axis.

Q.18 (3 Marks)

Evaluate $\int x^2/(x^2+1) dx$.

OR

Evaluate $\int x\sqrt{x^2+4} dx$.

Q.19 (3 Marks)

Find the equation of the line through $(1, 2, 3)$ parallel to $2i - j + 3k$.

OR

Find the angle between $\vec{a} = i + 2j + 2k$ and $\vec{b} = 2i + j - 2k$.

SECTION - D : LONG ANSWER QUESTIONS

Q.20 (4 Marks)

Find the area enclosed between $y = x^2$ and $y = 2x$.

OR

Evaluate $\int_0^{\pi/2} \sin x / (1 + \cos x) dx$.

Q.21 (4 Marks)

Solve $dy/dx + y \tan x = \sin x$.

OR

Solve $dy/dx - y/x = x^2$.

Q.22 (4 Marks)

A box contains 5 red, 4 blue and 3 green balls. Two balls are drawn at random without replacement. Find the probability that (i) both are red, (ii) one is red and the other is blue.

OR

A card is drawn from a well-shuffled pack of 52 cards. Find the probability that it is (i) a king, (ii) a red face card.

Q.23 (4 Marks)

Find the shortest distance between $r = (i + 2j - k) + \lambda(2i - j + k)$ and $r = (2i - j + 3k) + \mu(i + j - k)$.

OR

Find the equation of the plane through $(1, -2, 3)$ perpendicular to $2i - 3j + k$.

MARKS VERIFICATION

Question	Marks
Q.1	6
Q.2	6
Q.3	6
Q.4	7
Q.5	7
Q.6-Q.15	20
Q.16-Q.19	12
Q.20-Q.23	16
TOTAL	80

$6 + 6 + 6 + 7 + 7 + 20 + 12 + 16 = 80$ Marks

SHORT ANSWER KEY

Q.1: C, C, D, B, C, B. Q.2: equivalence; $\cos x$; 0; $e^{\int \tan x dx}$; 0; 0. Q.3: False, True, True, True, True, True. Q.4: (i)-(b), (ii)-(a), (iii)-(c), (iv)-(d), (v)-(e), (vi)-(f), (vii)-(g). Q.5: 2; 4; 10; 1; 0; 90° ; $1/(1+x^2)$.

For Q.6–Q.23, evaluate according to correct mathematical method, steps, formulae and final answer. Full solutions are intentionally not included.